

SOFTWARE ENGINEERING COURSE

LECTURE 9

Software Project Management and Planning

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MODULE 1 SCHEDULE

Week	Dates	Topic	SE Chapters
Week 9	Oct. 28–Nov1, 2024	Software Testing & Quality Management	Chapter 8 & 24
Week 10	Nov. 4– Nov 8, 2024	Software Evolution & Configuration Management	Chapter 9 & 25
Week 11	Nov. 11– Nov 15, 2024	Software Project Management & Software Project Planning	Chapter 22 & 23
Week 12	Nov. 18– Nov 22, 2024	Software Quality and Configuration	Chapter 24 & 25
Week 13	Nov. 25– Nov 29, 2024	Guest Lecture: Customer Success & Software Development	None
Week 14	December 2–6, 2023	Trends in Software Engineering	None





LEARNING OBJECTIVES

1. Risk management
2. Managing people
3. Plan-driven development
4. Agile planning





SOFTWARE PROJECT MANAGEMENT

- »» Concerned with activities involved in ensuring that software is delivered on time and on schedule and in accordance with the requirements of the organisations developing and procuring the software.
- »» Project management is needed because software development is always subject to budget and schedule constraints that are set by the organisation developing the software.





SUCCESS CRITERIA

- »» Deliver the software to the customer at the agreed time.
- »» Keep overall costs within budget.
- »» Deliver software that meets the customer's expectations.
- »» Maintain a coherent and well-functioning development team.





SOFTWARE MANAGEMENT DISTINCTIONS

- »» The product is intangible.
 - Software cannot be seen or touched. Software project managers cannot see progress by simply looking at the artefact that is being constructed.
- »» Many software projects are 'one-off' projects.
 - Large software projects are usually different in some ways from previous projects. Even managers who have lots of previous experience may find it difficult to anticipate problems.
- »» Software processes are variable and organization specific.
 - We still cannot reliably predict when a particular software process is likely to lead to development problems.





FACTORS INFLUENCING PROJECT MANAGEMENT

- »» Company size
- »» Software customers
- »» Software size
- »» Software type
- »» Organizational culture
- »» Software development processes
- »» These factors mean that project managers in different organizations may work in quite different ways.





MANAGEMENT ACTIVITIES

- **Risk management:** Project managers assess the risks that may affect a project, monitor these risks, and take action when problems arise.
- **People management:** Project managers have to choose people for their team and establish ways of working that lead to effective team performance.
- **Project planning:** Project managers are responsible for planning, estimating and scheduling project development and assigning people to tasks.
- **Reporting:** Project managers are usually responsible for reporting on the progress of a project to customers and to the managers of the company developing the software.
- **Proposal writing:** The first stage in a software project may involve writing a proposal to win a contract that describes the objectives of the project and how it will be carried out.





RISK MANAGEMENT

- Risk management is concerned with identifying risks and drawing up plans to minimise their effect on a project.
- Software risk management is important because of the inherent uncertainties in software development.
 - These uncertainties stem from loosely defined requirements, requirements changes due to changes in customer needs, difficulties in estimating the time and resources required for software development, and differences in individual skills.
- You have to anticipate risks, understand the impact of these risks on the project, the product and the business, and take steps to avoid these risks.





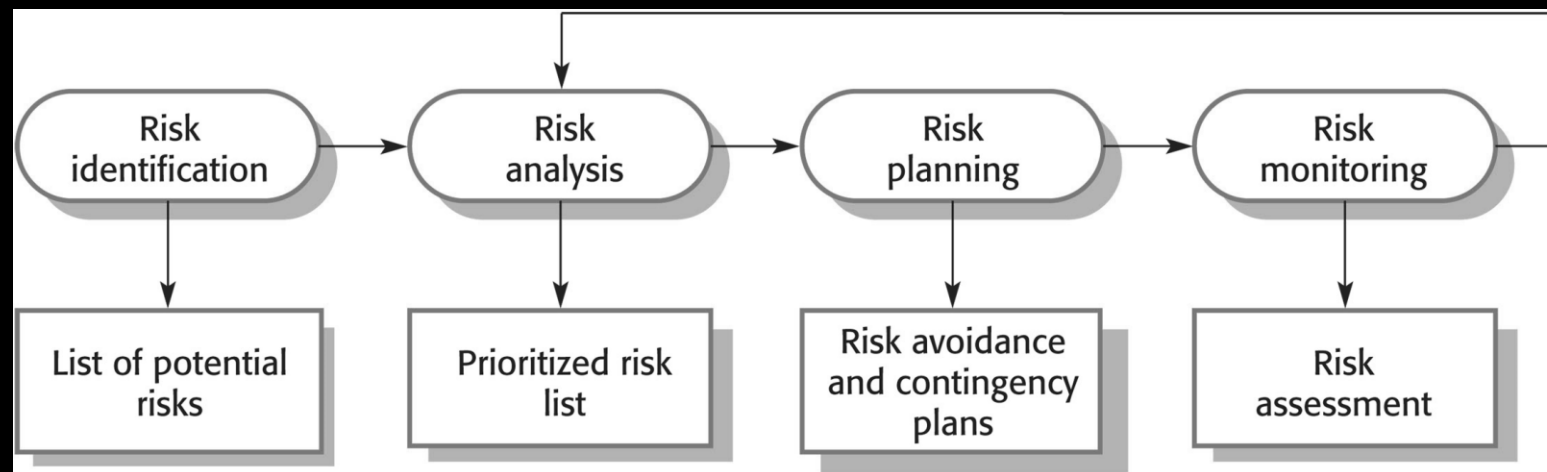
RISK CLASSIFICATION

- »» There are two dimensions of risk classification
 - The type of risk (technical, organizational, ..)
 - What is affected by the risk
- »» **Project risks** affect schedule or resources;
- »» **Product risks** affect the quality or performance of the software being developed;
- »» **Business risks** affect the organisation developing or procuring the software.



THE RISK MANAGEMENT PROCESS

- Risk identification: Identify project, product and business risks;
- Risk analysis: Assess the likelihood and consequences of these risks;
- Risk planning: Draw up plans to avoid or minimise the effects of the risk;
- Risk monitoring: Monitor the risks throughout the project;





RISK IDENTIFICATION

- May be a team activities or based on the individual project manager's experience.
- A checklist of common risks may be used to identify risks in a project
 - Technology risks.
 - Organizational risks.
 - People risks.
 - Requirements risks.
 - Estimation risks.



RISK ANALYSIS

- Assess probability and seriousness of each risk.
- Probability may be very low, low, moderate, high or very high.
- Risk consequences might be catastrophic, serious, tolerable or insignificant.





RISK PLANNING

- »» Consider each risk and develop a strategy to manage that risk.
- »» Avoidance strategies
 - The probability that the risk will arise is reduced;
- »» Minimization strategies
 - The impact of the risk on the project or product will be reduced;
- »» Contingency plans
 - If the risk arises, contingency plans are plans to deal with that risk;





WHAT-IF QUESTIONS

- »»What if several engineers fell ill at the same time?
- »»What if an economic downturn leads to budget cuts of 20% for the project?
- »»What if the performance of open-source software is inadequate and the only expert on that open-source software leaves?
- »»What if the company that supplies and maintains software components goes out of business?
- »»What if the customer fails to deliver the revised requirements as predicted?



RISK MONITORING

- »» Assess each identified risks regularly to decide whether or not it is becoming less or more probable.
- »» Also assess whether the effects of the risk have changed.
- »» Each key risk should be discussed at management progress meetings.





MANAGING PEOPLE

- »» People are an organisation's most important assets.
- »» The tasks of a manager are essentially people-oriented. Unless there is some understanding of people, management will be unsuccessful.
- »» Poor people management is an important contributor to project failure.





PEOPLE MANAGEMENT FACTORS

- **Consistency:** Team members should all be treated in a comparable way without favourites or discrimination.
- **Respect:** Different team members have different skills and these differences should be respected.
- **Inclusion:** Involve all team members and make sure that people's views are considered.
- **Honesty:** You should always be honest about what is going well and what is going badly in a project.





MOTIVATING PEOPLE

- An important role of a manager is to motivate the people working on a project.
- Motivation means organizing the work and the working environment to encourage people to work effectively.
 - If people are not motivated, they will not be interested in the work they are doing. They will work slowly, be more likely to make mistakes and will not contribute to the broader goals of the team or the organization.
- Motivation is a complex issue but it appears that there are different types of motivation

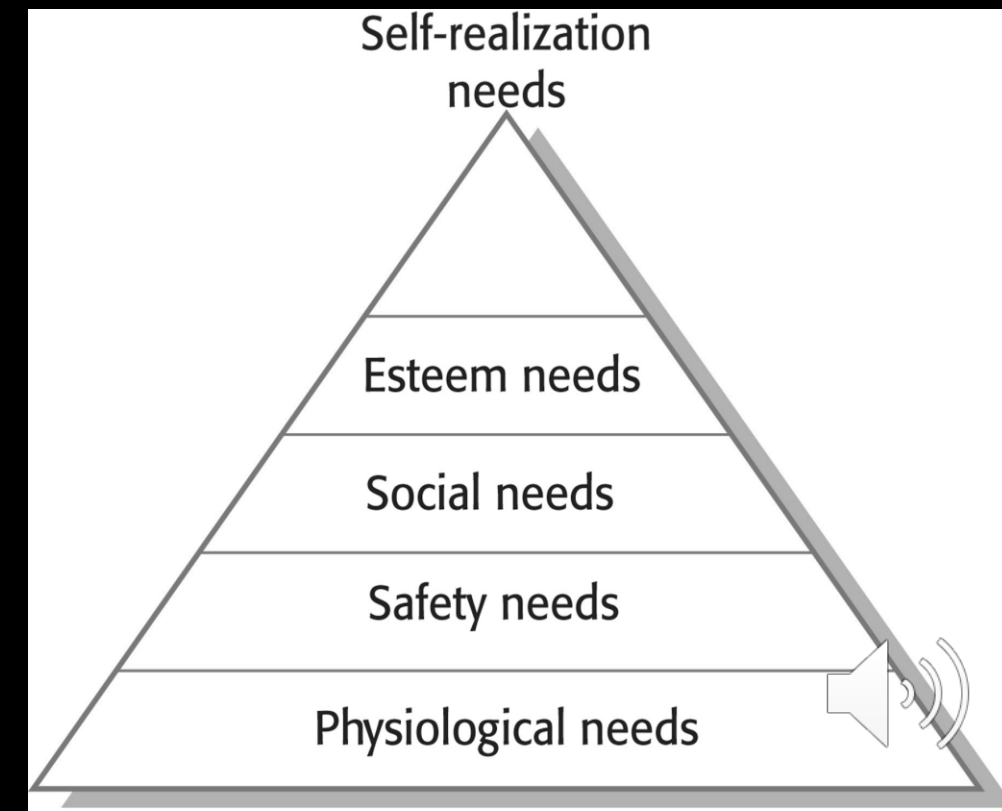


NEED SATISFACTION

- » In software development groups, basic physiological and safety needs are not an issue.
- » Social
 - Provide communal facilities;
 - Allow informal communications
- » Esteem
 - Recognition of achievements;
 - Appropriate rewards.
- » Self-realization
 - Training - people want to learn more;
 - Responsibility.

(Maslow's Hierarchy of Needs)

theory that suggests human needs are arranged in a hierarchical order. Once lower-level needs are met, individuals strive to fulfill higher-level needs.





NEED SATISFACTION

This theory can be applied to project management to understand and motivate team members, create a positive work environment, and ultimately achieve project success.

» Physiological Needs:

- In the workplace, this translates to providing basic necessities like fair wages, comfortable working conditions, and adequate breaks.
- Project managers should ensure that team members have the resources they need to perform their tasks effectively, such as necessary tools, equipment, and technology.

» Safety Needs:

- This involves creating a secure and stable work environment.
- Project managers can contribute by providing clear expectations, consistent feedback, and a sense of job security. Also establish clear project plans, timelines, and risk management strategies to help alleviate team members' concerns.



NEED SATISFACTION

» Belonging:

- This level focuses on social connections and a sense of belonging.
- Project managers can foster a positive team culture by encouraging collaboration, open communication, and team-building activities. Recognizing and appreciating team members' contributions can also help them feel valued and connected to the project.

» Esteem Needs:

- This level involves the need for respect, recognition, and achievement.
- Project managers can meet this need by providing opportunities for growth and development, recognizing and rewarding achievements, and offering constructive feedback. Delegating tasks and responsibilities can also boost team members' self-esteem.

» Self-Actualization Needs:

- This highest level involves realizing one's full potential and pursuing personal goals.
- Project managers can support this by assigning challenging tasks, encouraging innovation, and providing opportunities for professional development. Creating a culture of continuous learning and improvement can also help team members reach their full potential.





PERSONALITY TYPES

- The needs hierarchy is almost certainly an over-simplification of motivation in practice.
- Motivation should also take into account different personality types:
 - Task-oriented people, who are motivated by the work they do. In software engineering.
 - Interaction-oriented people, who are motivated by the presence and actions of co-workers.
 - Self-oriented people, who are principally motivated by personal success and recognition.





PROJECT PLANNING

- »» Project planning involves breaking down the work into parts and assign these to project team members, anticipate problems that might arise and prepare tentative solutions to those problems.
- »» The project plan, which is created at the start of a project, is used to communicate how the work will be done to the project team and customers, and to help assess progress on the project.





PLANNING STAGES

- At the proposal stage, when you are bidding for a contract to develop or provide a software system.
- During the project startup phase, when you have to plan who will work on the project, how the project will be broken down into increments, how resources will be allocated across your company, etc.
- Periodically throughout the project, when you modify your plan in the light of experience gained and information from monitoring the progress of the work.





PROPOSAL PLANNING

- »» Planning may be necessary with only outline software requirements.
- »» The aim of planning at this stage is to provide information that will be used in setting a price for the system to customers.
- »» Project pricing involves estimating how much the software will cost to develop, taking factors such as staff costs, hardware costs, software costs, etc. into account





PROJECT STARTUP PLANNING

- At this stage, you know more about the system requirements but do not have design or implementation information
- Create a plan with enough detail to make decisions about the project budget and staffing.
 - This plan is the basis for project resource allocation
- The startup plan should also define project monitoring mechanisms
- A startup plan is still needed for agile development to allow resources to be allocated to the project





DEVELOPMENT PLANNING

- »» The project plan should be regularly amended as the project progresses and you know more about the software and its development
- »» The project schedule, cost-estimate and risks have to be regularly revised





PLAN-DRIVEN DEVELOPMENT

- Plan-driven or plan-based development is an approach to software engineering where the development process is planned in detail.
 - Plan-driven development is based on engineering project management techniques and is the 'traditional' way of managing large software development projects.
- A project plan is created that records the work to be done, who will do it, the development schedule and the work products.
- Managers use the plan to support project decision making and as a way of measuring progress.





PLAN-DRIVEN DEVELOPMENT – PROS AND CONS

- »» The arguments in favor of a plan-driven approach are that early planning allows organizational issues (availability of staff, other projects, etc.) to be closely taken into account, and that potential problems and dependencies are discovered before the project starts, rather than once the project is underway.
- »» The principal argument against plan-driven development is that many early decisions have to be revised because of changes to the environment in which the software is to be developed and used.





PROJECT PLANS

- In a plan-driven development project, a project plan sets out the resources available to the project, the work breakdown and a schedule for carrying out the work.
- Plan sections
 - Introduction
 - Project organization
 - Risk analysis
 - Hardware and software resource requirements
 - Work breakdown
 - Project schedule
 - Monitoring and reporting mechanisms



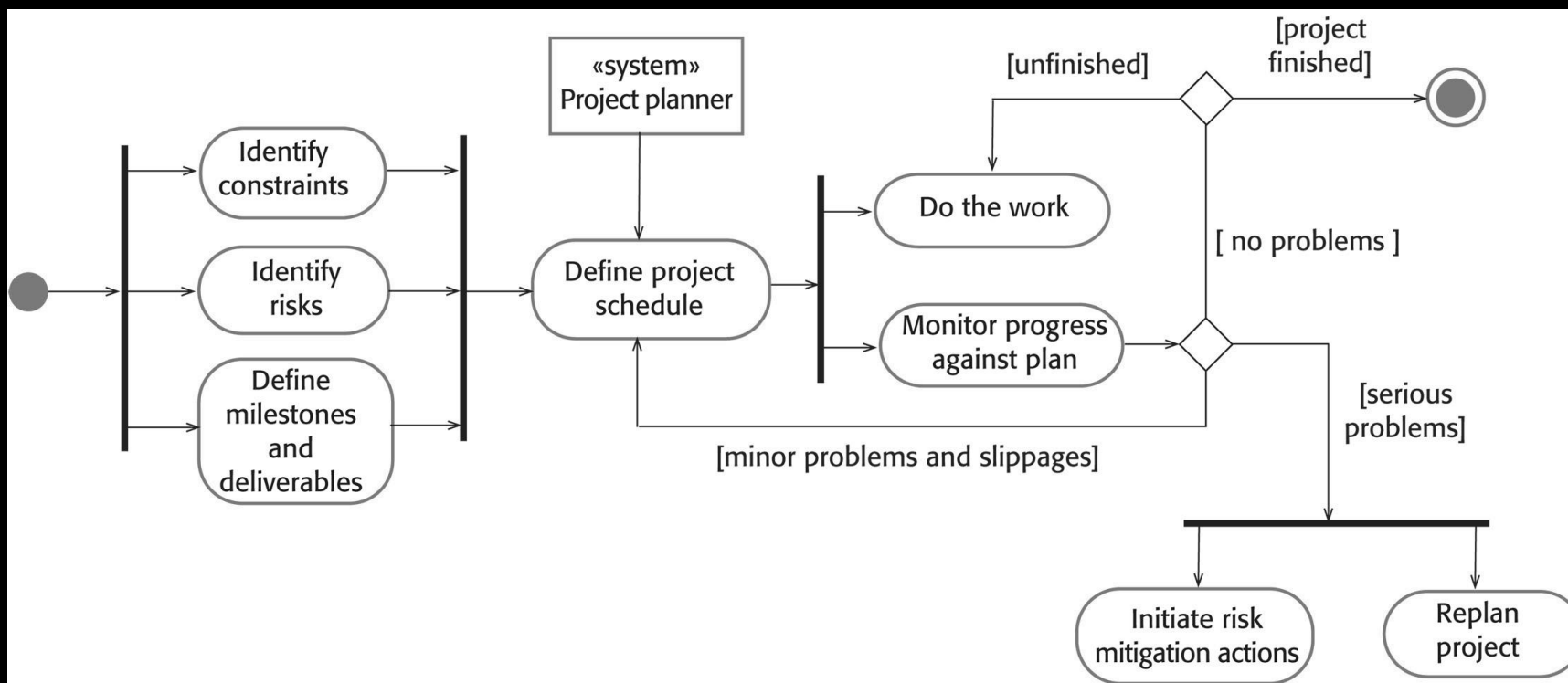


THE PLANNING PROCESS

- Project planning is an iterative process that starts when you create an initial project plan during the project startup phase.
- Plan changes are inevitable.
 - As more information about the system and the project team becomes available during the project, you should regularly revise the plan to reflect requirements, schedule and risk changes.
 - Changing business goals also leads to changes in project plans. As business goals change, this could affect all projects, which may then have to be re-planned.



THE PROJECT PLANNING PROCESS





PLANNING ASSUMPTIONS

- »» You should make realistic rather than optimistic assumptions when you are defining a project plan.
- »» Problems of some description always arise during a project, and these lead to project delays.
- »» Your initial assumptions and scheduling should therefore take unexpected problems into account.
- »» You should include contingency in your plan so that if things go wrong, then your delivery schedule is not seriously disrupted.





RISK MITIGATION

- If there are serious problems with the development work that are likely to lead to significant delays, you need to initiate risk mitigation actions to reduce the risks of project failure.
- In conjunction with these actions, you also have to re-plan the project.
- This may involve renegotiating the project constraints and deliverables with the customer. A new schedule of when work should be completed also has to be established and agreed with the customer.



AGILE PLANNING

- »» Agile methods of software development are iterative approaches where the software is developed and delivered to customers in increments.
- »» Unlike plan-driven approaches, the functionality of these increments is not planned in advance but is decided during the development.
 - The decision on what to include in an increment depends on progress and on the customer's priorities.
- »» The customer's priorities and requirements change so it makes sense to have a flexible plan that can accommodate these changes.





AGILE PLANNING STAGES

- Release planning, which looks ahead for several months and decides on the features that should be included in a release of a system.
- Iteration planning, which has a shorter term outlook, and focuses on planning the next increment of a system. This is typically 2-4 weeks of work for the team.





SOFTWARE DELIVERY

- »» A software increment is always delivered at the end of each project iteration.
- »» If the features to be included in the increment cannot be completed in the time allowed, the scope of the work is reduced.
- »» The delivery schedule is never extended.





AGILE PLANNING DIFFICULTIES

- »» Agile planning is reliant on customer involvement and availability.
- »» This can be difficult to arrange, as customer representatives sometimes have to prioritize other work and are not available for the planning game.
- »» Furthermore, some customers may be more familiar with traditional project plans and may find it difficult to engage in an agile planning process.





AGILE PLANNING APPLICABILITY

- »» Agile planning works well with small, stable development teams that can get together and discuss the stories to be implemented.
- »» However, where teams are large and/or geographically distributed, or when team membership changes frequently, it is practically impossible for everyone to be involved in the collaborative planning that is essential for agile project management.



